Neutron Diffusion Simulation Report

Samira Achilova

1 Introduction

Nuclear reactors rely on a self-sustaining chain reaction in which neutrons produced by fission induce further fission events. The spatial and temporal behavior of neutrons within a reactor core determines key operational characteristics such as power distribution, stability, and safety margins. Accurately predicting neutron behavior is therefore a central problem in reactor physics. In principle, neutron transport is described by the Boltzmann transport equation, which accounts for the direction, energy, and position of neutrons. However, this equation is high-dimensional and computationally expensive to solve. For many reactor-scale problems, it is sufficient to use the neutron diffusion approximation, which treats neutron motion as a diffusive process driven by spatial gradients in the neutron flux. Within this framework, neutrons undergo three primary processes:

- Production through fission reactions
- Absorption by materials in the system
- Leakage out of the system boundaries

The balance between these processes determines whether the system is subcritical, critical, or supercritical. This behavior is quantified by the effective multiplication factor $k_{\text{eff}} < 1$:

- $k_{\text{eff}} < 1$: subcritical
- $k_{\text{eff}} = 1$: critical
- $k_{\text{eff}} > 1$: supercritical

In this work, the neutron diffusion equation is solved numerically using finite difference methods in both one and two dimensions. The study investigates how geometry, system size, and the presence of a reflector influence the effective multiplication factor and the spatial distribution of neutron flux.

2 Governing Equation

The steady-state one-group neutron diffusion equation is:

$$-\nabla \cdot (D\nabla\phi) + \Sigma_a\phi = \frac{1}{k}\nu\Sigma_f\phi \tag{1}$$

where ϕ is the neutron flux, D is the diffusion coefficient, Σ_a is the absorption cross-section, Σ_f is the fission cross-section, and ν is the number of neutrons produced per fission. This equation forms an eigenvalue problem in which the largest eigenvalue corresponds to k_{eff}.

3 Numerical Method

The equation is solved using a finite difference discretization on a uniform grid. The resulting system is written in matrix form as:

$$B\phi = kA\phi \tag{2}$$

and solved using a generalized eigenvalue solver.
Key features of the numerical approach include:

- 1D tridiagonal Laplacian discretization
- 2D five-point stencil
- Harmonic averaging of diffusion coefficients across material interfaces
- Vacuum boundary conditions approximated using zero-flux treatment

4 Material Properties

The simulation uses homogenized U-235 fuel with the following properties:

- $D = 1.13$ cm
- $\Sigma_a = 0.106$ cm^{-1}
- $\Sigma_f = 0.0978$ cm^{-1}
- $\nu = 2.43$

The infinite-medium multiplication factor is:

$$k_\infty = \frac{\nu\Sigma_f}{\Sigma_a} \approx 2.24 \tag{3}$$

5 Results and Discussion

5.1 1D Bare Slab

The 1D simulation for a 150 cm slab yields:

$$k_{\text{eff}} = 2.2317 \tag{4}$$

The numerical flux agrees closely with the analytical cosine solution:

$$\phi(x) \propto \cos\left(\frac{\pi x}{a}\right) \tag{5}$$

This confirms the correctness of the numerical method. The high value of k_{eff} is expected due to minimal leakage in a large system.

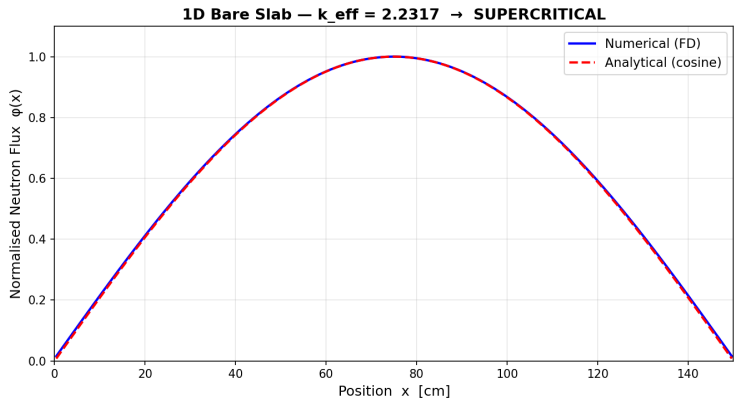

Figure 1: Numerical and analytical neutron flux in a 1D bare slab.

5.2 Criticality Sweep

A sweep over slab thickness shows that:

- Small systems are subcritical due to leakage
- The system becomes critical at approximately 9–10 cm
- Larger systems approach k_∞

This demonstrates the importance of geometry in determining reactor criticality.

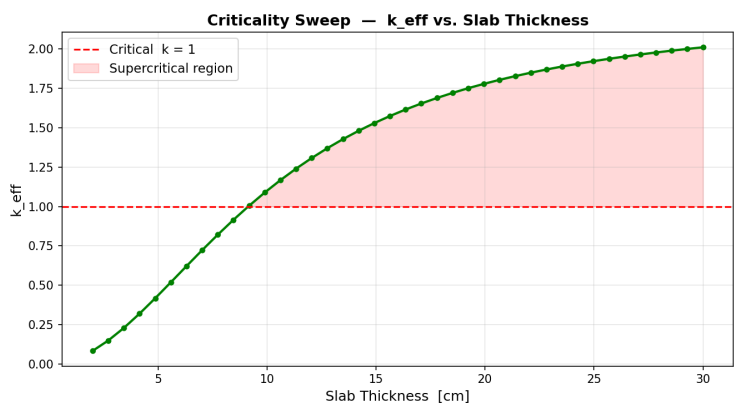

Figure 2: Effective multiplication factor as a function of slab thickness.

5.3 2D Bare Core

For a 40 cm × 40 cm bare fuel square, the result is:

$$k_{\text{eff}} = 1.9890 \tag{6}$$

The flux is symmetric and peaks at the center, decreasing toward the boundaries. This matches the expected analytical form:

$$\phi(x, y) \propto \cos\left(\frac{\pi x}{a}\right) \cos\left(\frac{\pi y}{a}\right) \tag{7}$$

The lower value of k_{eff} compared to 1D is due to increased leakage in two dimensions.

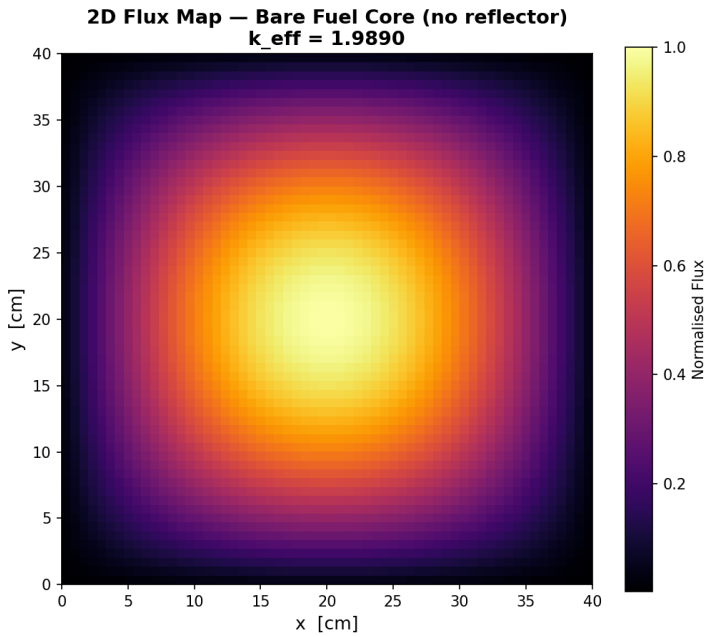

Figure 3: 2D neutron flux distribution for a bare fuel core.

5.4 2D Core with Reflector

Adding a reflector leads to a broader and flatter flux distribution. Neutrons that would otherwise escape are scattered back into the core, reducing leakage. This results in a higher effective multiplication factor and improved neutron economy.

6 Physical Insights

- Leakage is a dominant factor controlling criticality
- Higher-dimensional systems experience more leakage
- Reflectors improve neutron retention
- Fundamental flux modes are smooth and symmetric

7 Conclusion

The simulation successfully demonstrates neutron diffusion behavior in both one and two dimensions. The results agree with theoretical expectations and highlight the importance of geometry and boundary conditions in reactor physics.

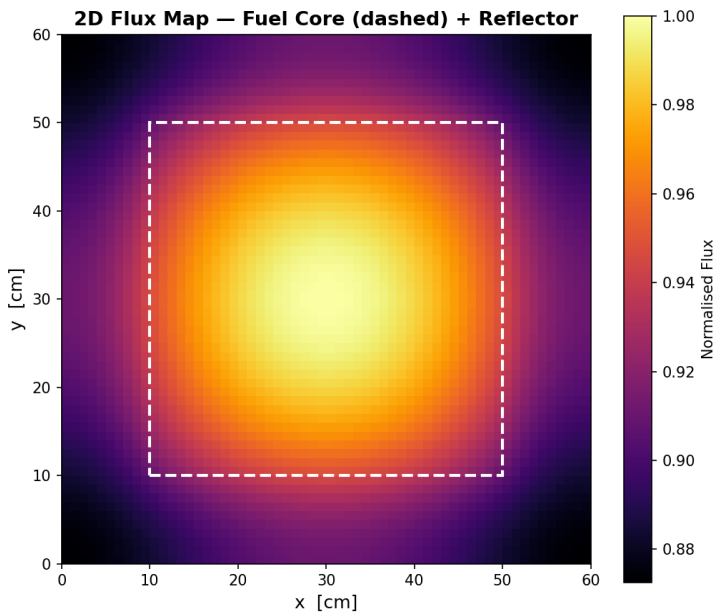

Figure 4: 2D neutron flux distribution for a fuel core with reflector.

8 Acknowledgement

Some parts of this work, including code debugging, numerical method clarification, and report structuring, were assisted by an AI tool (ChatGPT). All final interpretations and understanding of the results were verified independently.